

Review copy: September 7, 2026. Phase 1 is approved. This is the Phase 2 review module; its approval is pending.

2 Demand and Operating Profit

This section derives the commercial value kernel used by later manufacturing routes. It conditions on a project value shifter $q > 0$ and a positive marginal manufacturing cost $c > 0$. The route technologies that generate c are defined in the next section. Product price p is distinct from the qualified manufacturing-service price p_m .

Let C denote currency units and Y physical product-output units. One operating period has normalized length one. Both p and c are measured in C/Y , demand is measured in Y per period, one-period operating profit in C per period, and present value in C per commercially active product. A period profit is the total cash flow over that period, so discounting period profits does not sum an instantaneous flow without its time interval.

2.1 Residual demand and the pricing problem

Let $A > 0$ be the residual-demand scale and $\varepsilon > 1$ its constant price elasticity. For a given period, A has units $C^\varepsilon Y^{1-\varepsilon}$; q is dimensionless. The scalar A is distinct from the abandonment-route label. Residual demand for a successfully commercialized project is

$$y(p; q) = Aqp^{-\varepsilon}, \quad A > 0, \quad q > 0, \quad \varepsilon > 1. \quad (1)$$

The scale A , project shifter q , and elasticity ε are exogenous to the institutional regime. Conditional on q and c , the developer chooses product price from $p \in [c, \infty)$:

$$\max_{p \geq c} (p - c)Aqp^{-\varepsilon}. \quad (2)$$

Holding A, q, c, ε fixed, the derivative with respect to the product price is

$$\begin{aligned} \frac{\partial}{\partial p} [(p - c)Aqp^{-\varepsilon}] &= Aqp^{-\varepsilon} - \varepsilon Aq(p - c)p^{-\varepsilon-1} \\ &= Aqp^{-\varepsilon-1} [\varepsilon c - (\varepsilon - 1)p]. \end{aligned} \quad (3)$$

Since $Aqp^{-\varepsilon-1} > 0$, the derivative has the sign of the bracketed affine term. That term crosses zero exactly once, giving

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c. \quad (4)$$

Because $\varepsilon > 1$, $p^*(c) > c$, so the solution is feasible and interior.

For an independent second-order check, differentiate (3). At the stationary point, the term multiplying the derivative of $p^{-\varepsilon-1}$ vanishes, leaving

$$\left. \frac{\partial^2}{\partial p^2} [(p - c)Aqp^{-\varepsilon}] \right|_{p=p^*(c)} = -Aq(\varepsilon - 1)[p^*(c)]^{-\varepsilon-1} < 0. \quad (5)$$

The solution is also the unique global maximum. At the lower boundary $p = c$, profit is zero and the derivative is positive. The derivative is positive below $p^*(c)$ and negative above it, while profit converges to zero as p tends to infinity. Hence there is no competing boundary maximum.

2.2 Optimized one-period operating profit

Let $\pi(q, c)$ denote optimized one-period operating profit. Substituting (4) into the objective gives

$$\begin{aligned}\pi(q, c) &= \left(\frac{c}{\varepsilon - 1}\right) Aq \left(\frac{\varepsilon c}{\varepsilon - 1}\right)^{-\varepsilon} \\ &= Aq \frac{(\varepsilon - 1)^{\varepsilon-1}}{\varepsilon^\varepsilon} c^{1-\varepsilon}.\end{aligned}\tag{6}$$

The units are consistent: the price–cost margin is in C/Y, demand is in Y per period, and their product is in C per period.

Holding A, ε , and the non-differentiated argument fixed,

$$\frac{\partial \pi(q, c)}{\partial q} = \frac{\pi(q, c)}{q} > 0, \quad \frac{\partial \pi(q, c)}{\partial c} = (1 - \varepsilon) \frac{\pi(q, c)}{c} < 0.\tag{7}$$

The cost elasticity of optimized operating profit is $1 - \varepsilon$. These are continuous conditional derivatives in q and c , not a derivative with respect to the binary institutional regime.

2.3 Present value conditional on commercialization

Let $\beta \in (0, 1)$ be the one-period discount factor. Conditional on an already commercialized product, let $\varphi \in [0, 1)$ be the probability that it remains commercially active for one more period. The parameter φ is not the downstream realization probability $s(q)$: the latter applies before the operating stream begins and enters the route values defined below.

Let $t = 0, 1, \dots$ index operating periods, with the first commercial payment at $t = 0$, and let $R(q, c)$ denote expected operating present value conditional on commercialization. With a stationary conditional operating profit, this value is the convergent geometric sum

$$\begin{aligned}R(q, c) &= \sum_{t=0}^{\infty} (\beta\varphi)^t \pi(q, c) \\ &= \frac{\pi(q, c)}{1 - \beta\varphi}, \quad 0 \leq \beta\varphi < 1.\end{aligned}\tag{8}$$

Consequently, holding $\beta, \varphi, A, \varepsilon$ and the non-differentiated argument fixed,

$$\frac{\partial R(q, c)}{\partial q} = \frac{R(q, c)}{q} > 0, \quad \frac{\partial R(q, c)}{\partial c} = (1 - \varepsilon) \frac{R(q, c)}{c} < 0.\tag{9}$$

2.4 Boundary and accounting checks

The commercial-value kernel satisfies the following checks:

1. As $q \downarrow 0$, both $\pi(q, c)$ and $R(q, c)$ converge to zero.
2. As $c \rightarrow \infty$, both values converge to zero because $1 - \varepsilon < 0$.
3. If $\varphi = 0$, the commercial stream lasts one period in expectation and $R(q, c) = \pi(q, c)$.
4. As $\varphi \uparrow 1$, $R(q, c) \rightarrow \pi(q, c)/(1 - \beta)$, which remains finite because $\beta < 1$. The endpoint is a limiting check, not an admissible value under the frozen primitive domain $\varphi \in [0, 1)$.

5. The restriction $\varepsilon > 1$ is necessary for a finite interior markup price under this residual-demand specification. At $\varepsilon = 1$, profit approaches Aq as $p \rightarrow \infty$ but has no finite maximizer. For $0 < \varepsilon < 1$, profit is unbounded as $p \rightarrow \infty$.
6. As $c \downarrow 0$ at fixed $q > 0$, optimized profit diverges because $1 - \varepsilon < 0$. Thus $c > 0$ is a substantive domain restriction, not a removable endpoint. These boundary exercises do not enlarge the maintained parameter domain.

The object $R(q, c)$ is gross operating present value. The marginal cost c is subtracted exactly once through the operating margin $p - c$. Route-specific fixed organizational costs, the institutional burden on route E , and any separately defined qualified-capacity payment are absent from R ; later route-value equations must specify whether each manufacturing cost is embedded in a route-specific marginal cost or subtracted separately, and must not do both. Reduced-form return shortcuts are inactive in the baseline; the model uses only the explicit chain $y \rightarrow \pi \rightarrow R$.

Nothing in this section gives MAH a direct effect on A , q , ε , β , or φ . Thus this section establishes a policy-invariant commercial value kernel, not a policy comparative static.

The economic object here is operating revenue net of variable manufacturing cost, conditional on a project reaching commercialization. With elastic residual demand, the developer balances a higher margin against reduced sales. Higher marginal manufacturing cost raises the optimal product price and reduces the resulting operating-profit opportunity. Later organizational choices compare this opportunity with distinct setup, institutional and capacity costs. The present calculation alone neither selects a route nor predicts a change in project advancement or observed product counts.